

Gireeshee Pendela

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EDUCATION

Rutgers University, New Brunswick, NJ, USA

Master of Science, Data Science, GPA 3.57

2024-E. May 2026

Relevant Coursework: Database and Management Systems, Natural Language Processing, Statistical Learning

Indian Institute of Information Technology and Management, MP, India

Bachelor of Technology, Computer Science, GPA 7.31

2024

TECHNICAL SKILLS

Languages & Libraries: Python, SQL, JavaScript, HTML5, CSS3, NumPy, pandas, scikit-learn, NLTK, Matplotlib, Seaborn, R

Frameworks & Tools: Flask, Bootstrap 5, Jinja2, Git, GitHub, VS Code, Jupyter Notebook, Quarto, WordCloud

Databases: SQLite

Technical Areas: Natural Language Processing, Feature Engineering, Data Preprocessing, Data Visualization, Model Evaluation

RESEARCH PROJECTS

Hospital Management System

2025

Tech Stack: Python | Flask | SQLite | Bootstrap 5 | JavaScript | HTML/CSS | Jinja2

- **Expanded system functionality**, measured by **32+ role-specific application routes**, by designing end-to-end workflows for **3 distinct user roles** (Admin, Doctor, Patient)
- **Improved data integrity and automation**, measured by **12+ normalized tables and 10+ database triggers**, by enforcing relational constraints and automated validation across patient, appointment, and billing records
- **Reduced frontend code duplication by ~40%**, measured across **17+ dynamic pages**, by implementing reusable templates and shared layout components

Fake News Detection using NLP and Machine Learning

2025

Tech Stack: Python | scikit-learn | NLTK | pandas | NumPy | TF-IDF | Matplotlib | Seaborn | Jupyter Notebook

- Engineered an NLP preprocessing and feature pipeline, producing **218,495 TF-IDF features** from **44,000+ news articles** using **NLTK**
- Trained and benchmarked two binary classifiers, with **Logistic Regression achieving 98.45% accuracy (98% precision/recall)** on **8,854 test samples**, and **Multinomial Naive Bayes achieving ~96% accuracy**
- Evaluated and presented results using **5+ visualizations** (confusion matrices, feature importance, word clouds) and published **interactive Quarto documentation on GitHub Pages**

PROFESSIONAL EXPERIENCE

Project Intern | Matchday AI Technologies | India

2023

- **Implemented a full-scale web application using ReactJS**, reducing page load time by **~20%** through optimized component rendering and asset handling
- **Built a dynamic front-end dashboard for real-time sports analytics**, improving data visualization clarity and increasing **user engagement by ~40%**, measured via interaction frequency and dashboard usage

Project Intern | FluentGrid Limited | India

2023

- Built an end-to-end web application using **ReactJS, Angular, Bootstrap, and CSS3**, supporting **1,000+ users** and improving front-end performance
- Integrated **AMI** to optimize application workflows, achieving a **15% reduction in processing time** within a **3-month development cycle**

HONORS AND ACTIVITIES

Worked as a Grader for Data Wrangling

2025

WomenTech Network (Session monitor and facilitator)

2022

AI Planet Python Certification

2022

GirlScript Summer of Code

2022

Google Cloud Facilitator Program Certification

2021